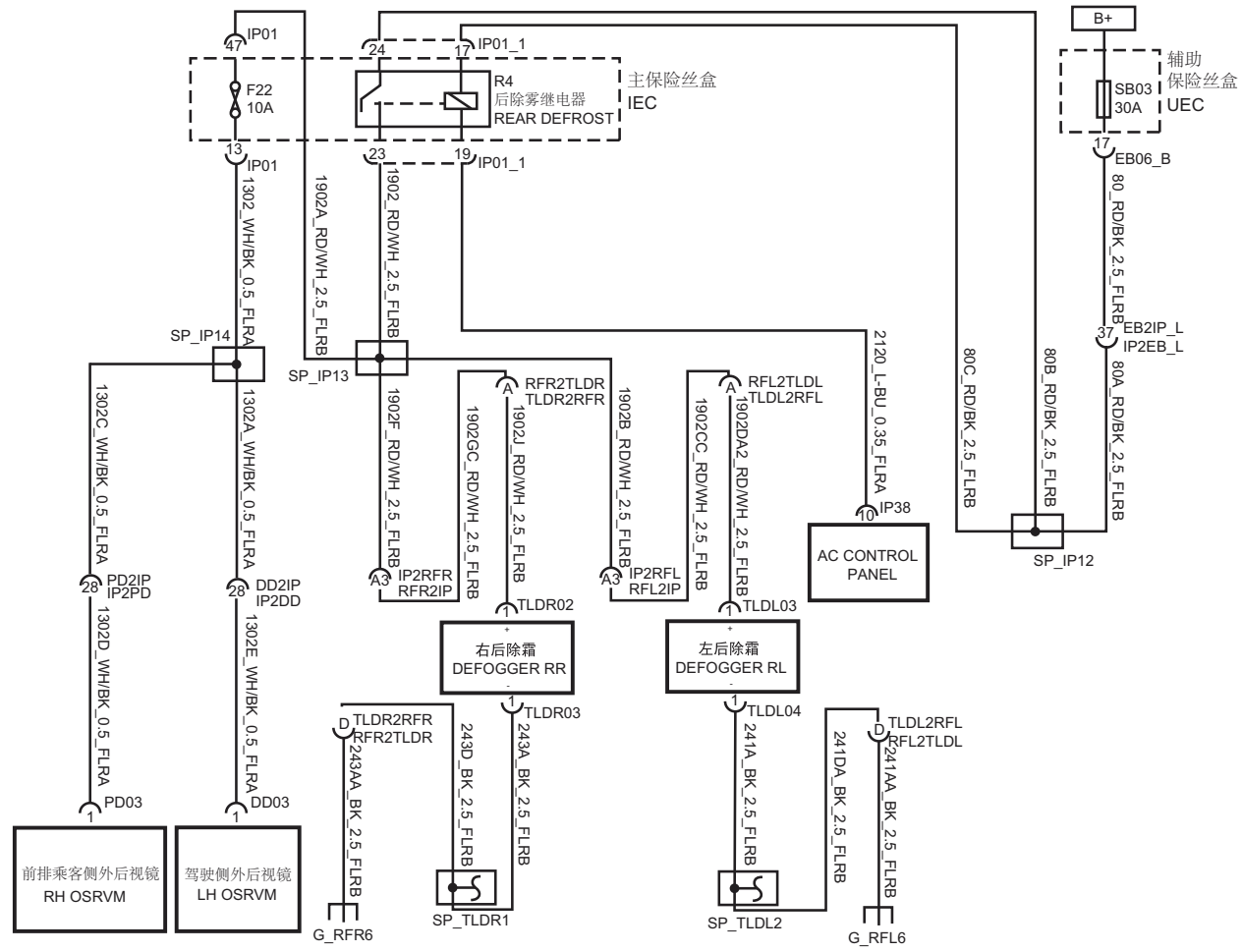
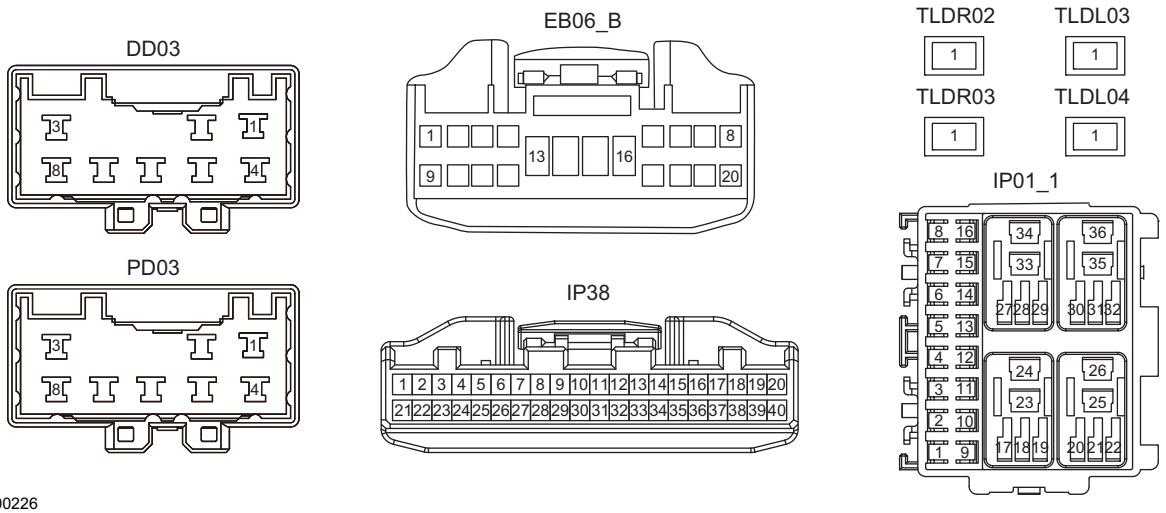
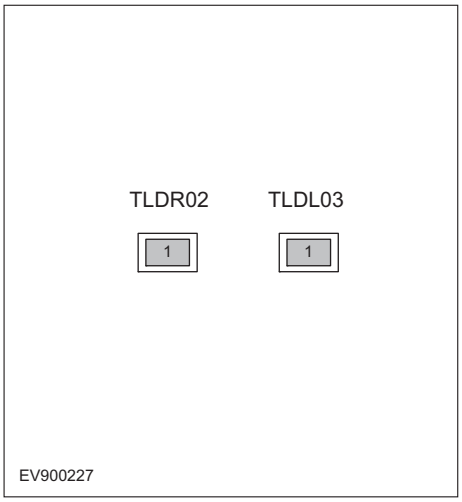
B1617-11、 B1617-12、 B1617-13

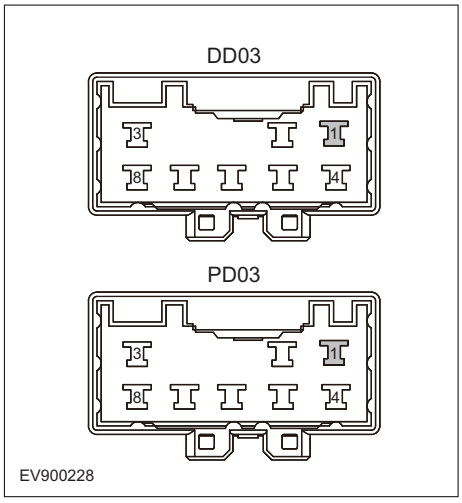
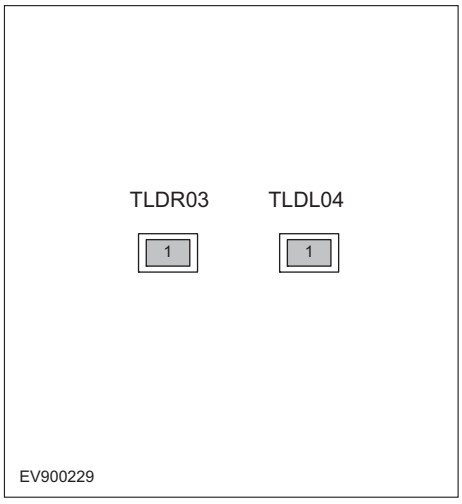
Fault diagnosis code
B1617-11：Rear window and mirror heating circuit short to ground
B1617-12：Rear window and mirror heating circuit short to battery
B1617-13：Rear window and mirror heating circuit open
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM Module fault • Relay fault • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1617-11
Rear window and mirror heating circuit short to ground
B1617-12
Rear window and mirror heating circuit short to battery
B1617-13
Rear window and mirror heating circuit open
To set the effect of a fault code condition
Rear window and mirror heating fault
Description of circuit diagram

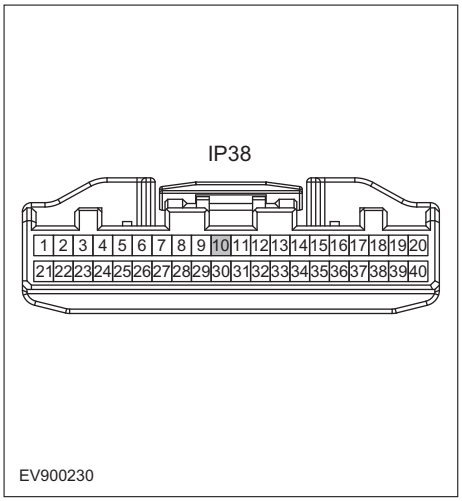
Circuit diagram

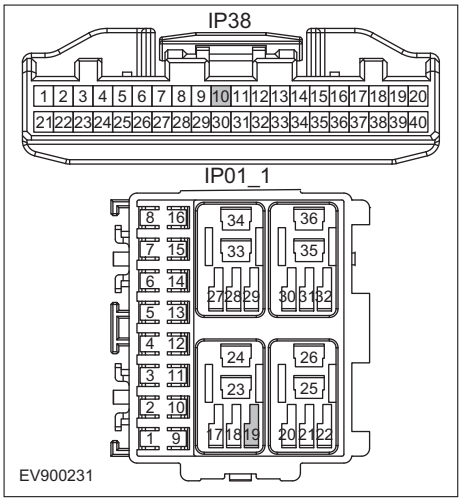




Troubleshooting steps		
Step 1	General inspection.	
		1 Check fuses SB03/F22 for damage. 2 Check the relay R4 for damage. 3 Check battery capacity. 1 Check the harness plug of rear window/mirror, BCM for damage, poor contact, aging and looseness. 2 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BCM with diagnostic apparatus. 4 Is there any other fault code except for B1617-11/B1617-12/B1617-13?
		YES → Refer to:DTC Summary list(BCM).
NO ↓		
Step 3	Check the supply power of rear window defrost when it is started or operating.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug TLDL03 of left rear defrost. 3 Disconnect the harness plug TLDR02 of right rear defrost. 4 Turn the ignition switch to ON. 5 Push the rear defrost switch. 6 Measure the voltage between the harness terminal of rear window defrost and ground. Measuring circuit: voltage between terminal 1 on plug TLDL03 and ground. Measuring circuit: voltage between terminal 1 on plug TLDR02 and ground. **Standard value:10~14V** 7 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 4	Check the supply power of rear view mirror heating when it is started or operating.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug DD03 of rear view mirror. 3 Disconnect the harness plug PD03 of side outside rear view of front row passenger. 4 Turn the ignition switch to ON. 5 Push the rear defrost switch. 6 Measure the voltage between the harness terminal of rear view mirror heating and ground. Measuring circuit: voltage between terminal 1 on plug DD03 and ground. Measuring circuit: voltage between terminal 1 on plug PD03 and ground. **Standard value:10~14V** 7 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Test whether the grounding circuit of rear window defrost is open.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug TLDL04 of left rear defrost. 3 Disconnect the harness plug TLDR03 of right rear defrost. 4 Measure the resistance between the harness terminal of horn and ground. Measuring circuit: resistance between terminal 1 on plug TLDL04 and ground. Measuring circuit: resistance between terminal 1 on plug TLDR03 and ground. **Standard value: < 1Ω** 5 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the control circuit of rear defrost (short circuit).
	1 Disconnect the harness plug IP38 of air conditioning control panel. 2 Measure the resistance between harness plug of air conditioning control panel and ground. Measuring circuit: resistance between terminal 10 on plug IP38 and ground. **Standard value: ∞** 3 Measure the voltage between harness plug of air conditioning control panel and ground. Measuring circuit: voltage between terminal 10 on plug IP38 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Check the control circuit of rear defrost (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP01_1 of IEC. 3 Measure the resistance between the harness terminal of IEC and air conditioning control panel. Measuring circuit: resistance between terminal 19 on plug IP01_1 and terminal 10 on plug IP38. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Replace the relay of rear defrost in good condition.
	1 Replace the relay of rear defrost in good condition. 2 Reconnect all disconnected wiring harness terminals. 3 Connect the fault diagnosis instrument to the diagnosis interface. 4 Turn the ignition switch to ON position. 5 Use the diagnostic apparatus to read BCM and DTC. 6 Clear the BCM and DTC. 7 Turn the ignition switch to OFF. 8 Waiting at least 15 seconds to allow the module to shut down properly. 9 Turn the ignition switch to ON again. 10 Use the diagnostic apparatus to read BCM and DTC again. 11 Test whether the faulted code is existed.
	NO → Replace the relay.
YES ↓	
Step 9	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the BCM Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	